

Waste materials

ANIMALS AND PLANTS USE A VARIETY OF METHODS TO GET RID OF THEIR WASTE MATERIALS.

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Excretion is the process of removing the waste produced by living bodies. This process is different to defecation, which is the release of the unused portion of food from the digestive tract.

Waste removal

A waste product is anything that the body cannot use. If they are allowed to build up in the body, they may become toxic. Nitrogen compounds from unneeded proteins form poisons that must be flushed away, and even carbon dioxide from respiration would make the blood dangerously acidic if it were not removed.

▽ Getting rid of waste

Organisms tackle their waste in different ways. The methods used to dispose of it safely depend on the nature of the waste and what resources are available. For example, fish flush waste out in water, but this method would dehydrate many animals, so other techniques are used.

REAL WORLD

Crocodile tears



The term “crying crocodile tears,” meaning someone acting sad without actually being upset, has a ring of truth to it. Crocodiles do indeed cry, but their tears are not emotional ones. The tears carry away unwanted salts from the body.

Waste product	Organism	Excretory process	Explanation
ammonia	fish	break-down of proteins	ammonia is very poisonous, so it is excreted in very dilute urine by fish and other animals that have plenty of water available around them
urea	mammals	break-down of proteins	to save water, animals chemically convert ammonia into urea, which is soluble and can be excreted in liquid urine
uric acid	birds, reptiles	break-down of proteins	uric acid is a solid form of nitrogen-containing waste excreted as a white paste, which saves water but requires a lot of energy to process
carbon dioxide	all life	respiration of sugars	carbon dioxide, produced as a byproduct of respiration, is released from the body during gas exchange, for example, in the lungs or gills
oxygen	plants and algae	photosynthesis	although oxygen is useful, too much can upset some of the plant's processes, so unwanted oxygen is released through its leaves
feces	most animals	undigested food	unneeded food material, combined with other waste materials (including brown pigments from dead blood cells), is eliminated via the anus
salt	all organisms	balancing concentrations of body fluids	salts help with many body processes, but too much can cause cramps and dehydration, so it is excreted in sweat, urine, or through skin glands